Viet Nguyen

+84.971.161.827 | nguyenhoangviet.hsgs@gmail.com | Viet Nguyen | github.com/hsgsstudent

Summary

My research lies at the intersection of statistics and machine learning, where I seek to understand the behavior of machine learning models from a rigorous theoretical perspective. My recent work focuses on parameter convergence rates in Mixture-of-Experts models. I am also interested in quantum computing, especially quantum optimization and quantum machine learning. My long-term goal is to develop statistical methodologies that enhance the reliability and interpretability of machine learning systems across both classical and quantum computing platforms.

Education

Hanoi University of Science and Technology (HUST) — Hanoi, Vietnam
BSc in Computer Science (Talented Program) — CPA: 3.9/4.0, Top 1% of the cohort — *2022 – 2026*

High School for Gifted Students, Hanoi University of Science (HUS) — Hanoi, Vietnam
High School Diploma (Specialized in Mathematics) — *2019 – 2022*

Experience

Independent Researcher — Oct 2025 – Present
Statistical Machine Learning Group (remote) — *Hanoi, Vietnam*

- Supervisor: Assoc Prof. Nhat Ho, The University of Texas at Austin
- Focused on the theoretical analysis of Mixture-of-Experts models
- Led two research projects resulting in submissions to NeurIPS 2026, currently under review.

Trivita AI (remote) — *Hanoi, Vietnam*

- Supervisor: Assoc Prof. Nhat Ho, The University of Texas at Austin
- Build AI model for medical image retrieval, focusing on colorectal cancer detection
- Fine-tuned DINOv2 with global search and local reranking, achieving the highest mAP among foundation models

Undergraduate Research Assistant — Jan 2024 – Sep 2025
Hanoi University of Science and Technology — *Hanoi, Vietnam*

- Supervisor: Dr. Trinh Chien, Hanoi University of Science and Technology
- Formulated the communication channel estimation problem using rigorous mathematical analysis
- Introduced an evolutionary algorithm to optimize channel quality up to 33%
- Led to a paper accepted by IEEE Transactions on Vehicular Technology

Pusan National University (remote) — *Hanoi, Vietnam*

- Supervisor: Dr. Trinh Chien, Hanoi University of Science and Technology
- Using quantum computer to solve multi-objective optimization problems
- Developed a quantum algorithm that optimized up to 9× faster than state-of-the-art classical methods
- Led to a paper accepted by IEEE Transactions on Quantum Engineering

Journal Publications

Advanced Quantum Annealing for the Bi-Objective Traveling Thief Problem: An ε-Constraint-Based Approach
Viet Nguyen, Nguyen Xuan Tung, Trinh Van Chien, Won-joo Hwang
IEEE Transactions on Quantum Engineering, 2026

Improved Differential Evolution for Enhancing the Aggregated Channel Estimation of RIS-Aided Cell-Free Massive MIMO
Trinh Van Chien, **Viet Nguyen**, Symeon Chatzinotas, Lajos Hanzo
IEEE Transactions on Vehicular Technology, 2025

Extending network lifespan and securing effective barrier coverage in 3D underwater wireless sensor networks
Nguyen Thi My Binh, Le Duc Anh Vu, **Nguyen Hoang Viet**, Huynh Thi Thanh Binh, Ho Viet Duc Luong
Cluster Computing, 2026

Conference Publications

A Mixture of Experts Vision Transformer for High-Fidelity Surface Code Decoding

Viet Nguyen, Manh Hung Nguyen, Hoang Ta, Van Khu Vu, Yeow Meng Chee

IEEE International Symposium on Information Theory (ISIT) 2026

Preprints

Rethinking Multinomial Logistic Mixture of Experts with Sigmoid Gating Function

Tuan Minh Pham*, Thinh Cao*, **Viet Nguyen***, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**

Under Review at NeurIPS 2026

A Statistical Theory of Gated Attention through the Lens of Hierarchical Mixture of Experts

Viet Nguyen*, Tuan Minh Pham*, Thinh Cao*, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**

Under Review at NeurIPS 2026

Achievement

Academic Encouragement Scholarship (5 semesters) 2022 – 2026

Scholarships for students with excellent academic achievements at HUST

Best Presentation Award Sep 2025

SoICT Summer School on AGENTIC AI

- *Project name: First steps toward Evaluating RAG through Agent-based Attack and Defense*

Vietnam Math Competition for University student 2025

The most prestigious math competition for university students in Vietnam (First Prize – ranked 8th)

Vietnam Math Competition for University student 2023

The most prestigious math competition for university students in Vietnam (First Prize – ranked 4th)

Vietnam Mathematical Olympiad 2020

The most prestigious math competition for high school students in Vietnam (Second Prize – ranked 13th)

Language

Vietnam: Native
English: Full professional proficiency, IELTS 7.0

References

Prof. Nhat Ho
Associate Professor, The University of Texas at Austin

Trinh Chien
Assistant Professor, Hanoi University of Science and Technology

Hoang Ta
Assistant Professor, Hanoi University of Science and Technology